

BiodivPond pilot workplan

1. General information

1.1. Title

Biodiversity monitoring of ponds: evaluating the applicability of novel methods and identification of obstacles to wide use

1.2. Short title

BiodivPond

1.3. Category

Species sub-pilot

1.4. Biodiversa+ priorities

The main priority is genomic and genetic monitoring using eDNA techniques to survey biodiversity in ponds. Another key priority is to monitor bats that commonly prey on the swarms of aquatic insects. However, the pilot focuses on monitoring ponds, thus also covering the adjacent wetlands. The data collection will take place in protected areas and include monitoring of insects; these priorities are also mentioned below.

- **Genomic and genetic monitoring:** Applications of genetic approaches to biodiversity monitoring across levels of biodiversity: harmonisation and gaps
- **Wetlands:** Integrative biodiversity monitoring of European wetlands, including mires and peatlands
- **Insects:** Monitoring insect biodiversity, with a particular focus on butterflies and other pollinators
- **Bats:** Monitoring the status and trends of bats across Europe
- **Protected Areas:** Biodiversity monitoring within protected areas in terrestrial, freshwater and marine realms, including Natura 2000 sites

2. The problem to be addressed

2.1. Context and general aim

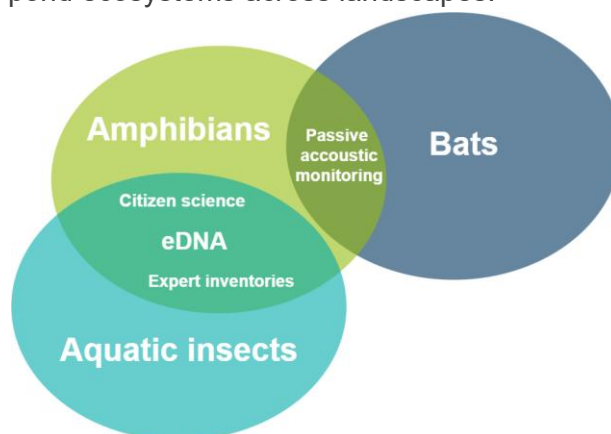
Freshwater biodiversity is under increasing threat globally (Reid et al., 2019) – particularly in Europe, where ecosystems face pressures from pollution, land-use intensification, habitat loss, and climate change (Zamora-Muñoz et al., 2024). In response, EU member states are prioritising freshwater restoration (EU, 2000; Regulation 2024/1991).

Among freshwater habitats, ponds are especially valuable. Though small (typically 1 m² to 5 ha – Richardson et al., 2022), they are the most numerous freshwater ecosystems on Earth and support a disproportionately high share of biodiversity, up to 70% of freshwater species in European landscapes (Williams et al., 2004; Biggs et al., 2017). Ponds also sustain terrestrial species – including bats – and provide vital ecosystem services such as water purification and carbon storage (Fehlinger et al., 2022; Holgerson & Raymond, 2016).

Despite their high importance, ponds remain largely neglected in EU water and nature strategies – including the Water Framework Directive – and are often missing from national monitoring schemes (Biggs et al., 2017; Vilmi et al., 2017). This is largely due to their high number and scattered distribution, which makes traditional monitoring of ponds logistically demanding and costly.

Innovative methods – including environmental DNA (eDNA) and passive acoustic monitoring (PAM) – offer scalable, cost-efficient alternatives. When combined with citizen science, these approaches enable high-resolution data collection while engaging local stakeholders, which is essential for long-term restoration success (Reid et al., 2019).

General aim: The sub-pilot aims to integrate pond monitoring approaches into a harmonised European initiative. By combining eDNA, PAM, and citizen-science-based methods, it will support the development of standardised, effective biodiversity monitoring schemes. This effort will fill critical knowledge gaps and help achieve the targets of the Nature Restoration Law through better data on pond ecosystems across landscapes.



2.2. Specific objectives

- 1) **Establish a transnational pond monitoring network across Europe:** Each active contributor will sample a minimum of 6 ponds (80 in total) during late spring of 2026 for biodiversity and key local environmental pond conditions following a highly

standardized sampling protocol based on traditional methods, eDNA and PAM, targeting amphibians, fish, aquatic macro-invertebrates, and bats.

- 2) **Design and pilot a citizen science-based biodiversity monitoring scheme for ponds across Europe:** All active contributors will collaborate in the development and implementation of a participatory citizen science-based monitoring scheme focusing on amphibians, fish and aquatic macro-invertebrates, actively building capacities to enable continuation of the scheme in the future. The monitoring will follow a standardised field protocol to assess key local environmental pond conditions and to collect samples for the analysis of eDNA. A total of 500 additional ponds is located in different biogeographical regions throughout Europe.
- 3) **Deliver a shared open-access data infrastructure:** A publicly accessible digital repository hosting all genomic (eDNA) and acoustic (PAM) data, including relevant metadata and species occurrence information, is operational by the project's end, facilitating transparency and reproducibility while utilising existing solutions.
- 4) **Operationalise data integration workflows:** A tested and transferable pipeline enabling the integration of eDNA and PAM data into national and international biodiversity databases, and Europe-wide frameworks is in place and usable by all project partners by the start of 2027 (for the core ponds) and further refined until the pilot's end.
- 5) **Provide user-friendly biodiversity visualisation tools:** Interactive tools and dashboards displaying species presence and geographical distributions have been developed and made available to stakeholders, including nature managers, policymakers, and the public, by the end of the pilot.

3. Relevance of the pilot study

3.1. Justification

Taxonomic focus: Monitoring targets amphibians, fish, aquatic macroinvertebrates (e.g. beetles, dragonflies, molluscs), and bats – key taxa for assessing pond biodiversity and restoration success.

Methodological approach: Combines two non-invasive, scalable techniques – environmental DNA (eDNA) for aquatic species, and passive acoustic monitoring (PAM) for bats – enabling standardised data collection across sites and countries.

Habitat focus: Temporary and permanent ponds located in different biogeographic regions of Europe. Ponds within protected areas (primarily Natura 2000) will be prioritised, but also unprotected sites will be assessed to capture relevant ecological gradients at the landscape scale.

Spatial design: Partners will cover representative pond types across multiple countries and regions, enabling transnational monitoring alignment. Inclusion of Natura 2000 sites by the active contributors will be encouraged in the site selection process. Synergies with the other Biodiversa+ Pilots (incl. ABMS, Soil, and Habitat Pilots) will be sought.

Policy relevance: Supports the implementation of the EU Biodiversity Strategy for 2030 and the Nature Restoration Regulation by generating restoration-relevant biodiversity data, especially for freshwater systems and connectivity indicators.

Links to existing frameworks: Collected data will contribute to several Essential Biodiversity Variables (EBVs) as outlined by EuropaBON. It will also feed into indicators relevant for bat activity and freshwater ecosystem condition.

Added value: The establishment of a cost-efficient monitoring network that supports evidence-based biodiversity policy, including in fragmented or data-poor regions.

3.2. European/transnational dimension

European biogeographical diversity will be represented through a transnational approach, with **biodiversity and key environmental conditions assessed across 580 ponds** that reflect major pond types found across the continent. Data on occurrence patterns will be collected, and the possibilities of an effective semi-quantitative census will be tested. Biodiversity monitoring delivers the greatest value when implemented beyond national borders using standardised methods and shared infrastructures. While technologies such as eDNA and passive acoustic monitoring can be applied at local scales, their broader utility lies in enabling coordinated, large-scale assessments, particularly within networks like Natura 2000 and the emerging Trans-European Nature Network (TEN-N).

This sub-pilot will show how harmonised biodiversity monitoring can be carried out effectively at a European scale. Through common protocols and centralised processing, the project will generate **interoperable data that supports EU-wide reporting**, restoration planning, and long-term species trend analysis. Collected data will feed into shared systems that enable cross-border validation, tool development, and consistent interpretation. This helps reduce duplication of effort and investment while strengthening accuracy and data coverage.

The approach supports the EU Biodiversity Strategy for 2030 and the Nature Restoration Regulation by providing Member States with the evidence and capacity needed to meet shared ecological restoration and monitoring goals. The recently adopted Water Resilience Strategy highlights the need to increase the natural water retention in the landscape, to reduce risks of flooding and drought, and increase natural groundwater recharge. Ponds and their adjacent wetlands are an important part of these nature-based solutions and provide synergy in biodiversity promotion. The pilot results will complement the existing monitoring required by the Water Framework Directive, which does not cover ponds frequently.

Species data will support the compilation of national and European Red List Assessments as well as the reporting according to the Habitats Directive.

3.3. Expected impact and future prospects

The sub-pilot will demonstrate how **genetic monitoring and sound recognition tools** can be integrated into biodiversity **monitoring frameworks for pond ecosystems**. By focusing on species that serve as indicators of ecosystem condition and resilience, the pilot will generate reliable, high-resolution data essential for planning, implementing, and evaluating restoration measures at the individual site level.

The pilot will directly support the legally binding targets of the **Nature Restoration Regulation**, which call for large-scale improvements in habitat quality and connectivity across member states. By offering scalable and cost-effective alternatives to traditional field methods, the pilot addresses one of the primary practical challenges in obtaining timely and accurate biodiversity data from ponds.

In addition to testing and refining these technologies, the **pilot will identify operational constraints to their wider adoption**, such as data processing capacity, interoperability with existing monitoring schemes, or site-level technical requirements, and propose solutions to overcome them. The resulting insights will support the development of targeted guidance and tools for practitioners and decision-makers across the EU.

All outputs will follow FAIR data principles. By piloting a replicable, genomic- and sensor-based approach to ecological monitoring, the initiative will contribute directly to the EU's ambition of **restoring degraded ecosystems and halting biodiversity loss** through informed, evidence-based action.

3.4. Other initiatives

- 1) **PONDERFUL**: Offers a strong scientific foundation for assessing pond biodiversity and climate resilience, including the eDNA surveys conducted.
- 2) **EuropaBON**: The pilot will contribute to EuropaBON's EBVs, particularly for freshwater and bat indicators, and align with its developing policy showcases, including those linked to the Birds and Habitats Directives.
- 3) **CIMaE**: This French alpine pond monitoring project provides opportunities to align eDNA protocols and expand the ecological gradient of monitored ponds. Especially relevant for sites in the Alpine biogeographic region.
- 4) **Biodiversa+ ABMS Pilot**: The pilot will build on the tools and workflows from the acoustic monitoring initiative to ensure compatibility and facilitate data integration. We will seek to improve the uptake of expert validation of samples to improve the PAM performance.

- 5) **ClimBats:** Collaboration will focus on using complementary acoustic data to understand climate impacts on bats and inform broader adaptation strategies. (<https://climbats.eu/>)
- 6) **GBIF:** Datasets will be available on the GBIF to ensure a shared open-access data infrastructure. We will follow the [recently published guidelines](#).

4. Implementation

4.1. Activities

Active contributors will implement a harmonised biodiversity monitoring scheme targeting amphibians, aquatic insects, and bats – taxa that are ecologically important but underrepresented in traditional conservation assessments. These multi-taxa approach leverages technological advances such as eDNA and passive acoustic monitoring alongside field-based inventories and citizen science, enabling efficient and scalable data collection. Site selection will focus on ponds and small waterbodies across a gradient of habitat conditions, ideally with existing conservation value or monitoring interest. At least 6 ponds per partner will be covered by the active contributors. An additional 30 ponds per partner will be covered by the citizen science scheme. Data from these sites will support the generation of Essential Biodiversity Variables (EBVs), including species presence/absence and diversity metrics.

Biodiversity data will be complemented with a minimal set of local environmental parameters, including pond nutrient status, vegetation coverage, surrounding land use, and climatic variability, to ensure that observed patterns in species data can be contextualised with environmental drivers.

Group	Number of sites	Amphibians and fish	Macroinvertebrates (Odonata, Coleoptera, Gastropoda)	Bats
Core ponds	80	eDNA (including quality check), <i>Triturus cristatus</i> abundance + traditional sampling	eDNA + traditional sampling	PAM
Citizen science ponds	500	eDNA	eDNA	-

Module 1 (core aquatic diversity): Amphibians, fish, and aquatic insects (Coleoptera, Odonata, Molluscs) will be monitored in 80 ponds through environmental DNA (eDNA) metabarcoding following highly standardised procedures. eDNA assessments are particularly suited to small waterbodies where traditional sampling is challenging. Each active contributor will collect eDNA samples from a selected set of ponds (minimum of 6) following standardised field protocols and chain-of-custody guidelines. Sampling design will be finalised based on the

active contributors needs as well as a simple methodology to assess vegetation coverage. Sample processing and analysis will be centralised with two partners to ensure consistency and standardisation. Data on key waterbody characteristics and surrounding land use will be collected to support the interpretation of observed biodiversity patterns.

Special attention will be directed towards the great crested newt (*Triturus cristatus*), a flagship species listed under Annex II of the Habitats Directive. We will assess its population density. *T. cristatus* from a subset of selected ponds using eDNA approaches and repeated traditional sampling. In Member States where *T. cristatus* is naturally absent, ecologically similar *Triturus* species will be targeted to enable cross-country comparisons and methodology harmonisation. The eDNA approach will also allow the detection of invasive species (e.g. *Xenopus* spp.), contributing to early warning efforts.

To ensure robustness of the methodology, a quality check and protocol testing exercise will be conducted among participating laboratories. This inter-lab comparison will involve testing identical eDNA samples across different labs using the agreed protocol, assessing consistency in DNA extraction, amplification, and species detection. This step is critical for ensuring interoperability, reproducibility, and building confidence in results across the partnership.

In parallel, expert teams will conduct site visits to validate habitat conditions and carry out *in situ* inventories on aquatic diversity. These conventional surveys will ground-truth the eDNA data and help generate reference records for improving bioinformatic pipelines. Sites where both eDNA and expert data are available will serve as testbeds for assessing cost-effectiveness, detection accuracy, and data complementarity of novel monitoring approaches.

Module 2 (acoustic monitoring): Each active contributor will deploy a network of Autonomous Recording Units (ARUs) to passively monitor bat activity and, where applicable, calling amphibians during the full growth season at the ponds sampled in Module 1. Devices such as AudioMoth or Song Meter Mini Bat will be used to record during peak activity periods. Acoustic recordings will be processed using automated pipelines, which assign species-level identifications alongside confidence scores.

This module addresses critical gaps in bat monitoring by providing systematic and scalable data on presence, activity levels, and seasonal dynamics. For amphibians, frog species with strong vocal signatures (e.g., *Pelophylax* spp., *Bombina* spp.) can be identified and monitored across the season. This will allow us to assess the overlap between genetic and acoustic monitoring techniques for selected taxa and evaluate the importance of small ponds for supporting bat populations.

Module 3 (citizen science scheme): In addition to the 80 ponds sampled by Module 1, an additional 500 ponds (minimum 30 per active contributor, depending on the final number of partners) will be monitored through citizen science, aiming to cover the different biogeographical regions within Europe. Citizen science will allow for the expansion of spatial

coverage of data collection and will foster community engagement through the participation of a diverse group of stakeholders. Outreach materials and training will be developed to ensure high data quality and safe fieldwork practices. National seminars will allow coordination, harmonisation of methods, and motivation of participants within each country.

4.2. Deliverables

Deliverable 1: Biodiversity monitoring report and method evaluation

A comprehensive report summarising biodiversity observations (amphibians, fish, aquatic insects, and bats) across all ponds. Includes:

- Comparison of eDNA, PAM, and traditional methods
- Results of the *Triturus cristatus* density estimation effort
- Evaluation of the inter-lab quality check (ring test)
- Recommendations for future use of these methods in routine biodiversity monitoring

Deliverable 2: Citizen science guideline

A practical guideline for designing and managing citizen science schemes focused on pond biodiversity monitoring using eDNA. It outlines a modular participation model adapted to different volunteer profiles, data quality protocols, recruitment and retention strategies, and recommendations for digital infrastructure.

Deliverable 3: *Triturus cristatus* monitoring report

A short technical report summarising the results of the density estimation of *T. cristatus* at core ponds. Includes insights into sampling effort, detection reliability via eDNA, and relevance for conservation status assessment.

Deliverable 4: Core pond dataset

A curated dataset with harmonised, quality-checked biodiversity records from the 80 core ponds sampled by active contributors. Includes both eDNA and acoustic data, as well as metadata on habitat conditions and pressures.

Deliverable 5: Citizen science pond dataset

A cleaned dataset containing biodiversity data collected from the 500 ponds sampled by the citizen scientists.

4.3. Timeline

The project will span 30 months from January 1st 2026, to June 30th 2027. The table estimates the workload for active contributors for each month of the project. Activities marked by an

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asterisk are partly or fully the responsibility of the coordinator institution and designated partners (Lab and PAM contributors), and the workload estimates are for active contributors only and are given in percent of the total workload.

Year	2026				2027				2028	
	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2
Core site selection										
Stakeholder site selection										
Pond sampling										
Lab protocols*										
Bioinformatics*										
PAM analysis*										
Data management and synthesis*						<u>D4</u>			<u>D5</u>	
National workshops										
Pilot contributor workshop										
Writing the reports*					<u>D3</u>				<u>D2</u>	<u>D1</u>

4.4. Partners involvement

Tasks of the **pilot coordinator** consist of:

- Having a steering role in the revision of the pilot action plan.
- Planning and monitoring the activities performed within the pilot, in collaboration with the general coordinator and active contributors.
- Having an overarching view of the pilot.
- Preparing (content-wise) and chairing the meetings of the pilot's working group.
- Reporting back on the activities of the pilot and drafting lessons learned from the pilot, including synthesising the main outcomes of the pilot, possibly using quantitative data if relevant;

- Reporting on the advancement of the activities to the Operational Team, for the activity report and other reporting due to the European Commission;
- Keeping a close collaboration with other pilot coordinators and with the general pilot coordinator;
- Facilitating the genomic analysis of the eDNA samples together with one more lab partner (see additional roles below)
- Ensuring consistency and links between the pilot activities and with other Biodiversa+ activities (for the latter point, support will be provided by the WP2 leader, the Biodiversa+ OT, as well as other Biodiversa+ task and sub-task leaders);
- Writing a data management plan for the biodiversity monitoring pilot. A first draft will be available at the latest two months after the launch of the pilot;
- Representing the pilot by possibly participating in events on behalf of Biodiversa+ and supporting the promotion of the pilot;
- Identifying relevant links with other initiatives and, when appropriate, supporting policy impact;
- Acting as a contact point with equipment providers or subcontractors related to the implementation of the activities of the whole pilot.

Tasks of the **active contributors** consist of:

- Providing feedback on the action plan of the pilot.
- Actively contributing to co-developing and co-implementing the activities of the pilot they are involved in at the level of Biodiversa+ and with the national relevant actors, including within the organisation of the Biodiversa+ partner, including supporting communication and outreach activities and answering relevant surveys;
- Reporting on the advancement of the activities to the pilot coordinator and the Operational Team for the activity report and other reporting due to the European Commission.
- Selecting sites for detailed sampling and coordinating the selection of ponds sampled by the stakeholders.
- Attending webinars to learn the methodologies for data collection.
- Providing input to the sampling design for data collection and the target species
- Deployment, service, and collection of data from bat ARUs.
- Collecting metadata at sensor deployment sites (vegetation survey, crown cover and microclimate measurements);
- Attending webinars and physical meetings to learn how to use acoustics recognisers for bat identification and in data interpretation;
- Involve bat species experts in the annotation of sounds collected and identified as target species for determining the optimal confidence score threshold of each species;
- Sending sound data to the pilot coordinator for analysis;
- Interpreting outcomes of monitoring campaigns and giving feedback on the reports;
- Coordinating the citizen science pond sampling in their country, storing the samples and sending them to the respective lab nodes;
- Organising the national workshops for the stakeholders involved in the citizen science scheme.

In addition, 3 individual roles need to be covered. Partners who will be selected for the position should demonstrate their expertise and show that they can deliver on time. In case of competition between partners, contributors need to come to an agreement or there has to be a formal competition with an anonymous vote among the active contributors will take place.

- **Two lab contributors** will complement the coordinating institution (NCA_CZ) in performing the laboratory and bioinformatic analyses of eDNA samples. One contributor will analyse the eDNA samples of vertebrates and the other of macroinvertebrates. Their tasks will include the extraction, purification, and sequencing of environmental DNA, as well as the development and application of bioinformatic pipelines for species identification. The lab contributor will process half of the samples collected from both the 80 core ponds and the 500 additional sites sampled with the support of citizen scientists. In addition, the lab contributor will estimate the abundance of *Triturus cristatus* at the 80 core ponds. Their contribution will ensure analytical consistency and high data quality across the entire sampling network.
- **The passive acoustic monitoring analysis contributor** will be responsible for the processing and interpretation of passive acoustic monitoring (PAM) data, with a primary focus on bat activity. They will co-develop data processing workflows and call recognition models, assist in calibrating automated detectors, and support data integration across regions. Together with the pilot coordinator, their role is central in ensuring that acoustic data collected across countries are processed in a standardised and interoperable way, enabling robust assessments of bat presence, activity patterns, and diversity.
- **Taxonomic analysis contributor** will conduct expert identification of aquatic macroinvertebrates collected using traditional sampling methods (e.g., kick nets, traps) from the 80 core ponds. The contributor will sort, identify, and catalogue specimens to species level where possible, following agreed standards. This work will serve as a critical reference for validating eDNA results and assessing the comparative effectiveness of traditional and molecular methods.

4.5. Coordinator quality

The coordinator of this sub-pilot is the **Nature Conservation Agency of the Czech Republic (NCA)**.

1) Coordination and chairing skills

- The Department of Biodiversity Monitoring at NCA has long-standing experience coordinating national and international biodiversity initiatives.
- Active institutional partner in multiple Biodiversa+ activities, including the ABMS Pilot and the Natura 2000 assessment coordination task.
- National lead for the development of a system to assess the conservation condition of species and habitats, including effectiveness indicators for Natura 2000 management.

- Coordinator of several national-scale LIFE projects (e.g. PROSPECTIVE LIFE), with roles in strategic planning, work package management, and multi-stakeholder engagement.
- Contributor to international conservation reporting obligations under the EU Habitats and Birds Directives, including membership in the EC Expert Group on Reporting on Nature Directives.
- Regular chair and organiser of expert working groups, including on amphibians, aquatic invertebrates, and bats, linking national data networks and field practice.

2) Topical expertise

- The department specialises in applied biodiversity monitoring, indicator development, and conservation data management.
- Leads or co-leads monitoring methodology development for over 200 species and 60 habitat types of European interest.
- Maintains institutional links with academic researchers, conservation NGOs, and regional authorities to ensure data uptake into management.
- Hosts national reference datasets for species distribution and habitat condition assessments.
- Organiser and participant in technical workshops and knowledge exchange seminars focused on nature-based solutions, pond restoration, eDNA, and acoustic monitoring.
- Publishes both peer-reviewed and grey literature on biodiversity indicators, field protocols, and evaluation methodologies, including materials developed under LIFE and OP Environment projects.

3) Resources available

- The department allocates dedicated staff to coordinate and technically support the BiodivPond pilot, including a coordinator and a data manager.
- NCA provides access to its biodiversity data infrastructure, expert networks, and laboratory partnerships.
- Internal collaboration between monitoring and GIS/IT units ensures efficient implementation of data pipelines, data sharing, and visualisation tools.

A qualified colleague at the NCA CZ will oversee day-to-day implementation, monitor overall project progress, and facilitate both internal and external communication among pilot partners and contributors. He will also lead the reporting activities and the synthesis of the main outcomes of the pilot. Jonáš Gaigr will be responsible for the coordination, utilising the experience from building a site condition evaluation framework within the Nature Conservation Information System, the ABMS Pilot, as well as other Biodiversa+ tasks.

4.6. Budget

Resources concerning personnel costs are allocated across pilot coordination and technical contributions to ensure effective implementation of the BiodivPond pilot. The exact PM rates and total personnel costs vary per partner according to national rates provided to the Biodiversa+ Operational Team.

- Pilot Coordination (NCA CR, Czechia):

The Nature Conservation Agency of the Czech Republic (NCA CR) acts as the coordinator. A total of 26 PM (104,000 €) is allocated.

This includes coordination, database development, and contribution to active tasks.

- Active Contributors:

Each participating partner is allocated 6 PM to sample and monitor the core ponds (cca 6 per partner) and coordinate the national citizen science sampling scheme. PM rates differ per partner (ranging approx. 2,000–9,200 € per PM).

- Lab Contributor (VL O):

A total of 5.5 PM (42,625 €) is allocated for eDNA sample analysis, covering laboratory work (extraction, QC, metabarcoding, etc.).

- Passive Acoustic Monitoring (SGAV):

3 PM (27,600 €) are allocated for acoustic data analysis.

- Taxonomic Analysis (NCA CR):

3 PM (12,000 €) are allocated for macroinvertebrate identification.

- Workshop Organisation (NCA CR, VL O):

Two workshops will be organised:

NCA CR: 0.5 PM (2,000 €)

VL O: 0.5 PM (3,875 €)

Each workshop includes 5,000 € (travel & subsistence) and 4,000 € (organisation of a 2-day workshop).

To support coordination and implementation, travel resources are allocated as follows:

- Pilot Coordinator (NCA CR):

A total of 8,800 € is allocated for participation in Biodiversa+ meetings, pilot workshops, and field visits.

- Active Contributors (per partner):
Standard allocation: 3,600 € (workshops + field visits).
Exception: MOE_FI – 4,600 € (higher field travel costs).

- Lab Contributor (VL O):
2,600 € for participation in pilot workshops.

The following costs relate to core biodiversity monitoring activities and sample processing:

- Active Contributors (per partner):

Equipment: 13,500 € (PAM equipment + aquatic sampling)
Other goods, works and services: 7,000 € (national workshops + sample shipping)

- Pilot Coordination (NCA CR):

Equipment: 13,500 €
Other goods, works and services: 13,000 € (incl. data infrastructure)

- Lab Contributor (VL O):

Equipment: 31,900 €
Other goods, works and services: 85,400 € (laboratory analysis + shipment)

- PAM Analysis (SGAV):

1,500 € for software and licenses

- Taxonomic Analysis (NCA CR):

2,000 € for chemicals and sample storage

5. Risk assessment

Risk: *Nagoya Protocol restrictions or unclear national ABS legislation may prevent the transfer of biological samples (e.g. eDNA) across borders.*

- **Likelihood:** Medium

- **Impact:** High
- **Mitigation:** A country-by-country Access and Benefit Sharing (ABS) compliance assessment is already underway. Sampling will occur only in countries with clear national frameworks and where Prior Informed Consent and Mutually Agreed Terms can be obtained on time. Partners will document all permits and declarations to meet EU due diligence obligations. No material will be collected from countries with unresolved legal status during the project timeframe.

Risk: *Delays in obtaining permits to sample ponds located in protected areas.*

- **Likelihood:** Medium
- **Impact:** Medium
- **Mitigation:** Site selection will prioritise locations with existing conservation activities and pre-established relationships with authorities. Local partners with experience in permit procedures will take the lead on applications. Where permitting is uncertain, alternative sites will be selected.

Risk: *No suitable partner may be identified to process the high volume of eDNA samples centrally in collaboration with the coordinator.*

- **Likelihood:** Medium
- **Impact:** High
- **Mitigation:** Identify multiple potential labs across countries; design modular analysis nodes; maintain flexibility to run samples nationally. The role can be covered or subcontracted by the pilot coordinator.

Risk: *Difficulty in finding an acoustic analysis partner. Lack of expertise or the availability of institutions capable of analysing passive acoustic data.*

- **Likelihood:** Medium
- **Impact:** Medium
- **Mitigation:** Engage early with acoustic experts; consider outsourcing part of the analysis or training new contributors. The role can be covered or subcontracted by the pilot coordinator.

Risk: *Limited stakeholder engagement may affect the coverage or frequency of sample collection in decentralised pond sites.*

- **Likelihood:** Medium
- **Impact:** Medium
- **Mitigation:** A citizen science approach is under development, including targeted training, accessible protocols, and regional coordinators. Engagement will be supported through regular communication, feedback loops, and co-developed incentives for contributors (e.g. access to data or visibility).

Progress will be monitored using a shared implementation calendar and milestone checklist aligned with deliverables. Regular coordination meetings (online and in-person) will assess risks dynamically and reallocate resources or adjust protocols as needed.

6. References

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Annex 1. Budget for BiodivPond pilot

Task	Partner acronym	PM rate	nb PM	PM (€)	PM description	Travel & subsistence	Equipment	Other goods, works and services	TOTAL Other direct costs	Description	TOTAL Subcontracting costs	Indirect costs (15%)	COVERED BY EC CONTRIBUTION
Workshop host	NCA CR	4,000.00 €	0.5	2,000.00 €	0.5PM to organise a workshop	5,000.00 €	- €	4,000.00 €	9,000.00 €	Travel & subsistence: 5,000 € to invite external experts. Other goods, works and services: 4000 € for a 2-day workshop.	- €	1,650.00 €	12,650.00 €
Workshop host	VLO	7,750.00 €	0.5	3,875.00 €	0.5PM to organise a workshop	5,000.00 €	- €	4,000.00 €	9,000.00 €	Travel & subsistence: 5,000 € to invite external experts. Other goods, works and services: 4000 € for a 2-day workshop.	- €	1,931.25 €	14,806.25 €
Active contributor in the whole pilot	BOZEN	5,500.00 €	6	33,000.00 €	6 PM for the duration of the pilot to sample and monitor the core ponds (cca 6 per partner) and coordinate the national citizen science sampling scheme	3,600.00 €	13,500.00 €	7,000.00 €	24,100.00 €	<u>Travel & subsistence:</u> 2,600€ to attend 2 pilot workshops (à 1,300€) and 1,000€ to go to field sites. <u>Equipment:</u> 12,300€ for the PAM equipment (SM5BAT, calibrator, solar power kit, SD cards, batteries and gear for installation) + 1,200€ for aquatic sampling. <u>Other goods, works and services:</u> 6,000€ to organise 2 national workshops for the stakeholders involved in the citizen science sampling scheme + 1000 € for shipping costs for eDNA samples.	- €	8,565.00 €	65,665.00 €
Active contributor in the whole pilot	ExEA	2,000.00 €	6	12,000.00 €	6 PM for the duration of the pilot to sample and monitor the core ponds (cca 6 per partner) and coordinate the national citizen science sampling scheme	3,600.00 €	13,500.00 €	7,000.00 €	24,100.00 €	<u>Travel & subsistence:</u> 2,600€ to attend 2 pilot workshops (à 1,300€) and 1,000€ to go to field sites. <u>Equipment:</u> 12,300€ for the PAM equipment (SM5BAT, calibrator, solar power kit, SD cards, batteries and gear for installation) + 1,200€ for aquatic sampling. <u>Other goods, works and services:</u> 6,000€ to organise 2 national workshops for the stakeholders involved in the citizen science sampling scheme + 1000 € for shipping costs for eDNA samples.	- €	5,415.00 €	41,515.00 €

Task	Partner acronym	PM rate	nb PM	PM (€)	PM description	Travel & subsistence	Equipment	Other goods, works and services	TOTAL Other direct costs	Description	TOTAL Subcontracting costs	Indirect costs (15%)	COVERED BY EC CONTRIBUTION
Active contributor in the whole pilot	MEPGT	3,173.21 €	6	19,039.26 €	6 PM for the duration of the pilot to sample and monitor the core ponds (cca 6 per partner) and coordinate the national citizen science sampling scheme	3,600.00 €	13,500.00 €	7,000.00 €	24,100.00 €	Travel & subsistence: 2,600€ to attend 2 pilot workshops (à 1,300€) and 1,000€ to go to field sites. Equipment: 12,300€ for the PAM equipment (SMSBAT, calibrator, solar power kit, SD cards, batteries and gear for installation) + 1,200€ for aquatic sampling. Other goods, works and services: 6,000€ to organise 2 national workshops for the stakeholders involved in the citizen science sampling scheme + 1000 € for shipping costs for eDNA samples.	- €	6,470.89 €	49,610.15 €
Active contributor in the whole pilot	MOEP	5,631.00 €	6	33,786.00 €	6 PM for the duration of the pilot to sample and monitor the core ponds (cca 6 per partner) and coordinate the national citizen science sampling scheme	3,600.00 €	13,500.00 €	7,000.00 €	24,100.00 €	Travel & subsistence: 2,600€ to attend 2 pilot workshops (à 1,300€) and 1,000€ to go to field sites. Equipment: 12,300€ for the PAM equipment (SMSBAT, calibrator, solar power kit, SD cards, batteries and gear for installation) + 1,200€ for aquatic sampling. Other goods, works and services: 6,000€ to organise 2 national workshops for the stakeholders involved in the citizen science sampling scheme + 1000 € for shipping costs for eDNA samples.	- €	8,682.90 €	66,568.90 €
Active contributor in the whole pilot	MOE_FI	8,500.00 €	6	51,000.00 €	6 PM for the duration of the pilot to sample and monitor the core ponds (cca 6 per partner) and coordinate the national citizen science sampling scheme	4,600.00 €	13,500.00 €	7,000.00 €	25,100.00 €	Travel & subsistence: 2,600€ to attend 2 pilot workshops (à 1,300€) and 2,000€ to go to field sites (field travel budget increased because of long distances). Equipment: 12,300€ for the PAM equipment (SMSBAT, calibrator, solar power kit, SD cards, batteries and gear for installation) + 1,200€ for aquatic sampling. Other goods, works and services: 6,000€ to organise 2 national workshops for the stakeholders involved in the citizen science sampling scheme + 1000 € for shipping costs for eDNA samples.	- €	11,415.00 €	87,515.00 €

Task	Partner acronym	PM rate	nb PM	PM (€)	PM description	Travel & subsistence	Equipment	Other goods, works and services	TOTAL Other direct costs	Description	TOTAL Subcontracting costs	Indirect costs (15%)	COVERED BY EC CONTRIBUTION
Active contributor in the whole pilot	SAS	4,000.00 €	6	24,000.00 €	6 PM for the duration of the pilot to sample and monitor the core ponds (cca 6 per partner) and coordinate the national citizen science sampling scheme	3,600.00 €	13,500.00 €	7,000.00 €	24,100.00 €	Travel & subsistence: 2,600€ to attend 2 pilot workshops (à 1,300€) and 1,000€ to go to field sites. Equipment: 12,300€ for the PAM equipment (SM5BAT, calibrator, solar power kit, SD cards, batteries and gear for installation) + 1,200€ for aquatic sampling. Other goods, works and services: 6,000€ to organise 2 national workshops for the stakeholders involved in the citizen science sampling scheme + 1000 € for shipping costs for eDNA samples.	- €	7,215.00 €	55,315.00 €
Active contributor in the whole pilot	SEPA	7,200.00 €	6	43,200.00 €	6 PM for the duration of the pilot to sample and monitor the core ponds (cca 6 per partner) and coordinate the national citizen science sampling scheme	3,600.00 €	13,500.00 €	7,000.00 €	24,100.00 €	Travel & subsistence: 2,600€ to attend 2 pilot workshops (à 1,300€) and 1,000€ to go to field sites. Equipment: 12,300€ for the PAM equipment (SM5BAT, calibrator, solar power kit, SD cards, batteries and gear for installation) + 1,200€ for aquatic sampling. Other goods, works and services: 6,000€ to organise 2 national workshops for the stakeholders involved in the citizen science sampling scheme + 1000 € for shipping costs for eDNA samples.	- €	10,095.00 €	77,395.00 €
Active contributor in the whole pilot	SGAV	9,200.00 €	6	55,200.00 €	6 PM for the duration of the pilot to sample and monitor the core ponds (cca 6 per partner) and coordinate the national citizen science sampling scheme	3,600.00 €	13,500.00 €	7,000.00 €	24,100.00 €	Travel & subsistence: 2,600€ to attend 2 pilot workshops (à 1,300€) and 1,000€ to go to field sites. Equipment: 12,300€ for the PAM equipment (SM5BAT, calibrator, solar power kit, SD cards, batteries and gear for installation) + 1,200€ for aquatic sampling. Other goods, works and services: 6,000€ to organise 2 national workshops for the stakeholders involved in the citizen science sampling scheme + 1000 € for shipping costs for eDNA samples.	- €	11,895.00 €	91,195.00 €

Task	Partner acronym	PM rate	nb PM	PM (€)	PM description	Travel & subsistence	Equipment	Other goods, works and services	TOTAL Other direct costs	Description	TOTAL Subcontracting costs	Indirect costs (15%)	COVERED BY EC CONTRIBUTION
Active contributor in the whole pilot	VLO	7,750.00 €	6	46,500.00 €	6 PM for the duration of the pilot to sample and monitor the core ponds (cca 6 per partner) and coordinate the national citizen science sampling scheme	3,600.00 €	13,500.00 €	7,000.00 €	24,100.00 €	<u>Travel & subsistence:</u> 2,600€ to attend 2 pilot workshops (á 1,300€) and 1,000€ to go to field sites. <u>Equipment:</u> 12,300€ for the PAM equipment (SM5BAT, calibrator, solar power kit, SD cards, batteries and gear for installation) + 1,200€ for aquatic sampling. <u>Other goods, works and services:</u> 6,000€ to organise 2 national workshops for the stakeholders involved in the citizen science sampling scheme + 1000 € for shipping costs for eDNA samples.	- €	10,590.00 €	81,190.00 €
BiodivPond pilot coordination (NCA_CZ)	NCA CR	4,000.00 €	26	104,000.00 €	4 PM per year for the coordination of the subpilot (coordinator), 4 PM per year to develop the database (data manager), 6 PM to fulfill the active contributor tasks	8,800.00 €	13,500.00 €	13,000.00 €	35,300.00 €	<u>Travel & subsistence:</u> 5 200 € for travels (á 1 300 €) to attend Biodiversa+ meetings (2 biodiversity monitoring weeks and Biodiversa+ final conference) and external conferences. An additional 2,600€ to attend 2 pilot workshops (á 1,300 €) and 1,000 € to go to field sites. <u>Equipment:</u> 12,300€ for the PAM equipment (SM5BAT, calibrator, solar power kit, SD cards, batteries and gear for installation) + 1,200 € for aquatic sampling. <u>Other goods, works and services:</u> 6,000 € to organise 2 national workshops for the stakeholders involved in the citizen science sampling scheme + 1000 € for shipping costs for eDNA samples + 6,000 € for developing the data infrastructure.	43,810.00 €	20,895.00 €	204,005.00 €
Lab contributor (vetrebrates)	VLO	7,750.00 €	5.5	42,625.00 €	5,5 PM to analyse the eDNA samples	2,600.00 €	31,900.00 €	85,400.00 €	119,900.00 €	<u>Travel & subsistence:</u> 2,600€ to attend 2 pilot workshops (á 1,300€) <u>Equipment:</u> 31,900€ for sampling kits & materials <u>Other goods, works and services:</u> 76,300€ for laboratory work (extraction & purification, QC, metabarcoding, ring test, crested newt ddPCR + IPC, and multimarker assay); 9,100 € for sample and aliquot shipment	- €	24,378.75 €	186,903.75 €
Passive Acoustic Monitoring analysis	SGAV	9,200.00 €	3	27,600.00 €	3 PM for the acoustic data analysis	- €	- €	1,500.00 €	1,500.00 €	Other goods, works and services: 1,500 € for software and licenses.	- €	4,365.00 €	33,465.00 €
Taxonomic analysis	NCA CR	4,000.00 €	3	12,000.00 €	3 PM for the identification of macroinvertebrates from the core ponds	- €	- €	2,000.00 €	2,000.00 €	Other goods, works and services: 2000 € for chemicals and storage of samples.	- €	2,100.00 €	16,100.00 €
			92.50	509,825.26 €		54,800.00 €	166,900.00 €	172,900.00 €	394,600.00 €		43,810.00 €	135,663.79 €	1,083,899.05 €